



TFAnalytics

Explainable market intelligence for the Team Fortress 2 economy

PROJECT SUMMARY

TFAnalytics collects and validates daily TF2 price-guide data, preserves each update as a historical snapshot, and turns that history into interactive dashboards, explainable market signals, item-level trends, and downloadable datasets. It helps players make better-informed decisions without promising profitable trades.

The idea: What inspired us to build it?

Team Fortress 2 has a long-running virtual economy in which cosmetics, weapons, taunts, and other items are valued in refined metal and Mann Co. Supply Crate Keys. Unusual items add another layer of complexity because the same hat can have many particle effects, each forming a distinct market. Thousands of non-Unusual variants are also differentiated by quality and craftability.

Community resources such as [backpack.tf](#) make individual price checks possible, but understanding how the market changes over time still requires players to inspect many entries and compare values manually. We wanted to complement those resources with a broader historical view of the economy.

This inspired TFAanalytics: one place where traders, collectors, and curious players can explore current values, compare past snapshots, and understand why a market is being highlighted. We present it as decision-support market intelligence rather than a guaranteed investment predictor.

How it works

Scheduled GitHub Actions workflows request daily pricing data from the [backpack.tf](#) API. One pipeline collects exact Unusual item-effect markets; the other collects the wider community price guide while excluding Unusual quality so the two datasets remain complementary. A separate Steam Web API job maintains the official item-image catalogue used by the website.

COLLECT	VALIDATE	ARCHIVE	EXPLORE
Daily price-guide and item-image data	Schemas, prices, duplicates, timestamps, row counts	Versioned CSVs and Google Sheets tabs	Streamlit pages, Plotly charts, and downloads

Python and pandas processing code cleans each snapshot and converts prices into consistent refined-metal and key equivalents. Automated validators check required columns, timestamps, price values, duplicates, and row count. Suspicious or malformed outputs fail the workflow instead of silently replacing reliable data.

Accepted raw and processed snapshots are committed to GitHub and copied to date-stamped Google Sheets tabs. The Streamlit website discovers the latest files, warns visitors if data becomes stale, and compares snapshots across time.

The interface uses Plotly for interactive visualisations. It includes headline metrics, item and effect leaderboards, historical trends, market sentiment, risk flags, opportunity screening, and confidence labels. Users can inspect an exact item variant and download the underlying CSV data.

What makes it stand out

TFAnalytics is more than a dashboard of today's values. It continually builds a reproducible history, protects that history with validation, and explains the evidence behind its signals. This combination of product, automation, and data quality makes the system useful even as the market changes.

Main features

- Automated daily collection covering exact Unusual markets and the wider non-Unusual price guide.
- Historical comparisons and searchable trends for items, qualities, and Unusual effects.
- Interactive Plotly charts, market summaries, and item and effect leaderboards.
- Explainable sentiment, opportunity, risk, and confidence indicators.
- Validation for missing fields, invalid prices, duplicates, timestamp mismatches, and suspicious data loss.
- Freshness warnings, versioned CSV history, Google Sheets archival, and website downloads.
- Cloud workflows that run without a team member's computer remaining online.

Technology stack

Technology	Role
Python	Collection, processing, validation, and analytics
Streamlit	Multi-page interactive website
pandas and NumPy	Cleaning, aggregation, comparison, and scoring
Plotly	Interactive visualisations and historical trends
Requests and backpack.tf API	Scheduled price-guide collection
Steam Web API	Official TF2 item images
Selenium and Beautiful Soup	Supplementary collection and parsing utilities
GitHub Actions	Scheduling, validation, archival, and CI
Google Sheets API	Date-stamped secondary archives
CSV and Git	Reproducible historical storage and version control

Intended audience

TFAnalytics is designed for TF2 traders, collectors, market researchers, and players who want to understand the economy before buying or selling an item. It helps them compare items and effects, monitor longer-term movement, and investigate unusual behaviour without examining thousands of entries. It also demonstrates how an automated pipeline can turn a changing virtual economy into a transparent analytics product. Prices are indicative community guide values, and all signals are evidence for further investigation, not financial advice.

Built by Isaac Foo, Koh Min Xuan, and Low Zong Xuan for the Build Beyond Hackathon.